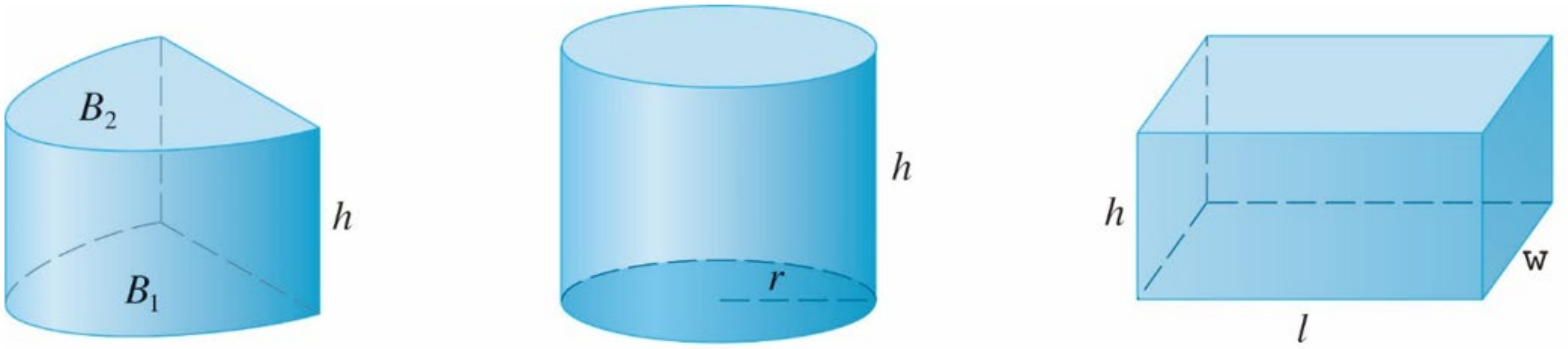
用截面積求體積

# 6-1 Volumes Using Cross-Sections

師大工教一

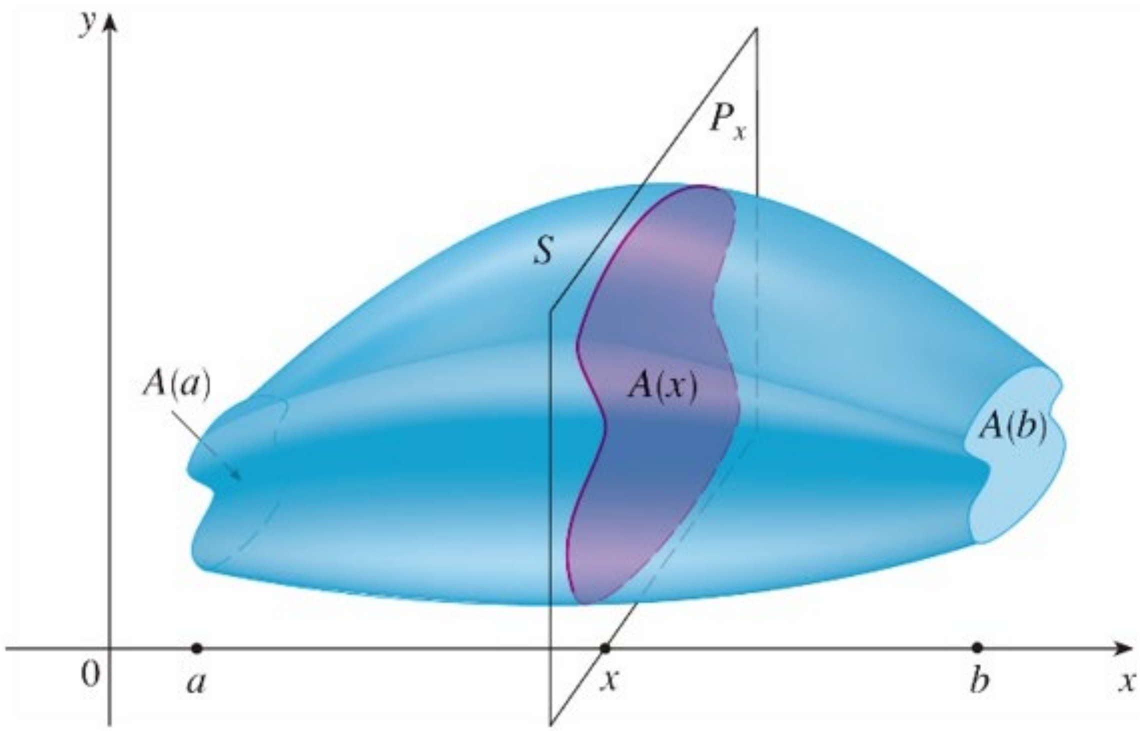
## Volumes of Cylinders

柱体

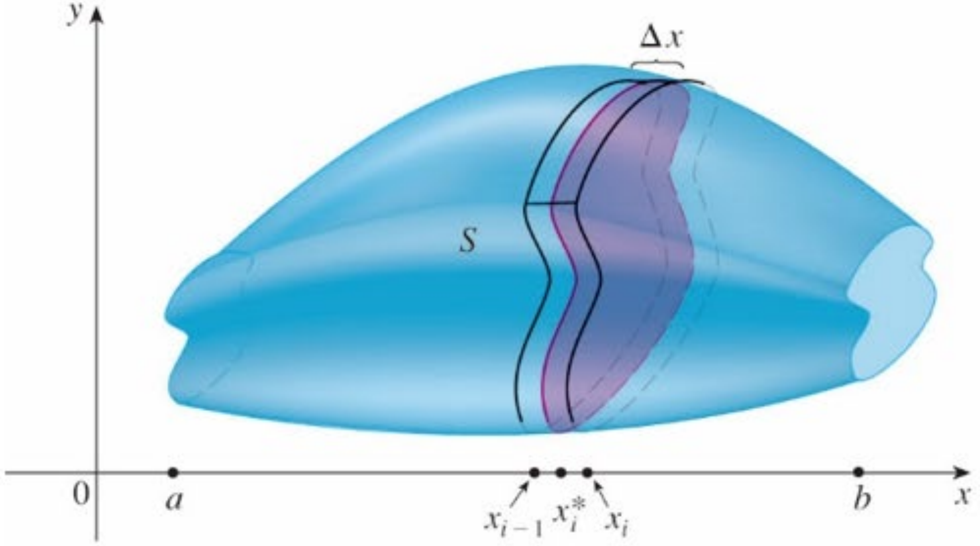


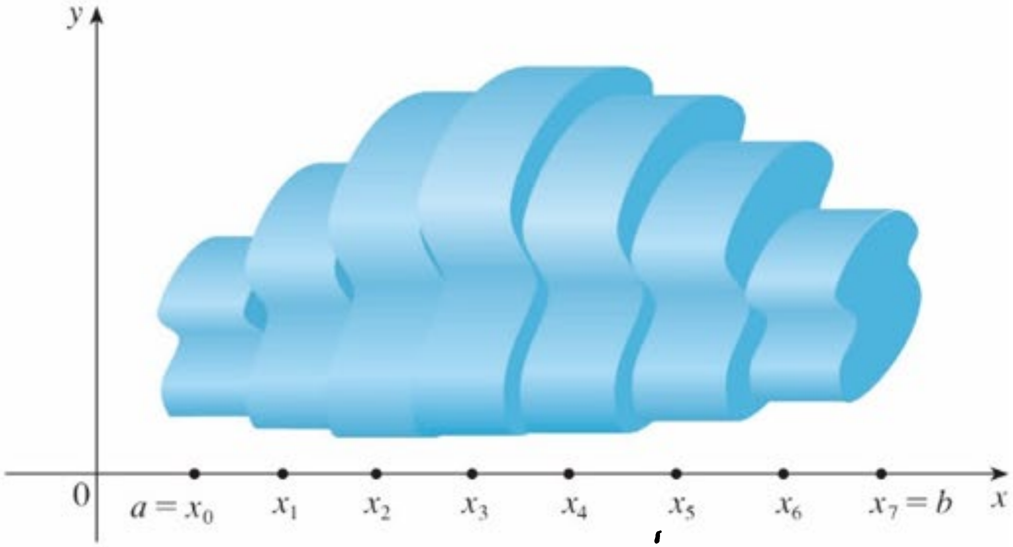

(a) Cylinder $V = Ah$ (b) Circular cylinder $V = \pi r^2 h$ (c) Rectangular box $V = lwh$

## Cross-sections ：(横)截面



# Method of Slicing




$$V \approx \sum_{k=1}^{n} V_k = \sum_{k=1}^{n} A(x_k)\,\Delta x_k$$

$$V = \int_a^b A(x)\,dx$$

$$\lim_{n\to\infty} \sum_{k=1}^{n} A(x_k)\,\Delta x_k = \int_a^b A(x)\,dx$$

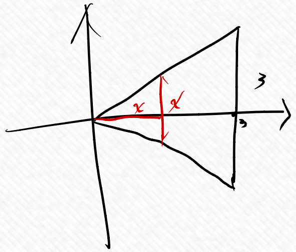
Definition The **volume** of a solid of integrable cross-sectional area $A(x)$ from $x = a$ to $x = b$ is the integral of $A$ from $a$ to $b$, $V = \int_a^b A(x)\,dx$.

**Calculating the Volume of a Solid**

1. Sketch the solid and a typical cross-section.
2. Find a formula for $A(x)$, the area of a typical cross-section.
3. Find the limits of integration.
4. Integrate $A(x)$ to find the volume.

金字塔

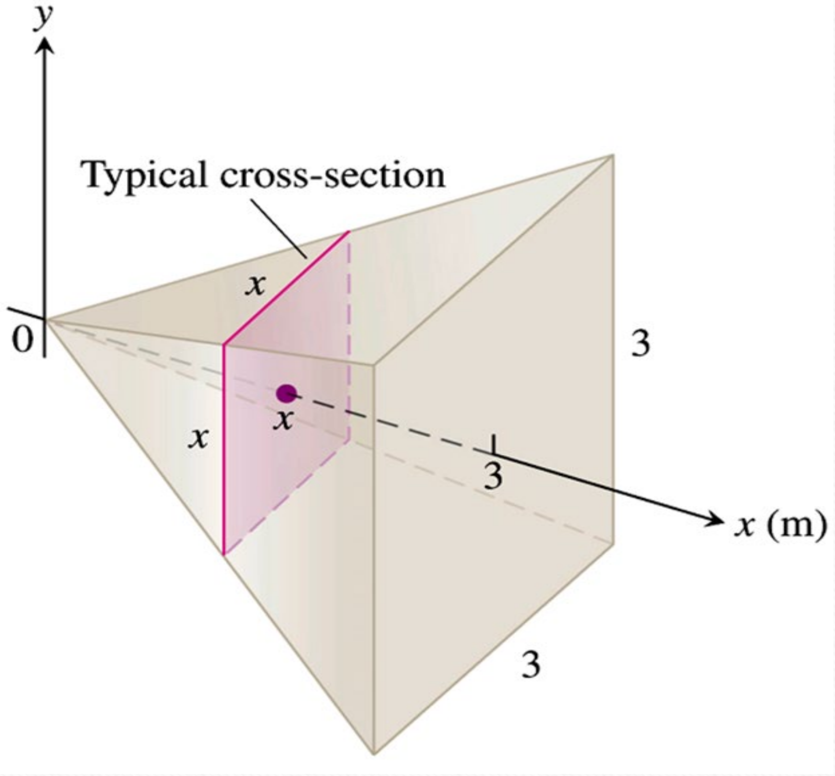
Ex1(p375) A pyramid 3 m high has a square base that is 3 m on a side. Find the volume of the pyramid.




锥体体積 $= \frac{1}{3}$ 柱体体積

$A = x^2$

$$V = \int_0^3 x^2\,dx$$
$$= \frac{x^3}{3}\Big|_0^3 = 9$$

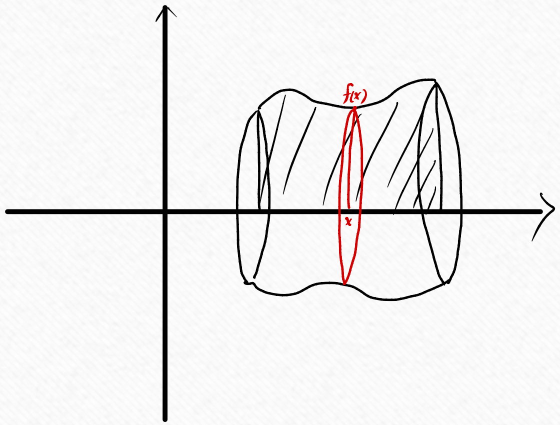


## Solid of Revolution: The Disk Method

Volume by Disks for Rotation About the $x-$Axis

$$V=\int_a^b A(x)\,dx=\int_a^b \pi\left[R(x)\right]^2 dx$$

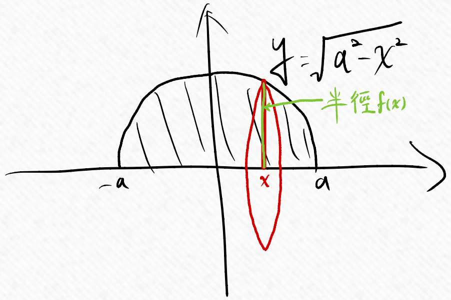
旋転体 : 圓盤法




$$V=\int_a^b A(x)\,dx=\int_a^b \pi\left(f(x)\right)^2 dx$$

Ex5(p377) The circle $x^2+y^2=a^2$ is rotated about the $x-$axis to generate a sphere. Find its volume.

球面




$x^2+y^2=a^2$

$y^2=a^2-x^2$

$y=\pm\sqrt{a^2-x^2}$

ball 球 半径a, $V=\frac{4}{3}\pi a^3$

$$V=\int_{-a}^{a}\pi\left(\sqrt{a^2-x^2}\right)^2dx$$

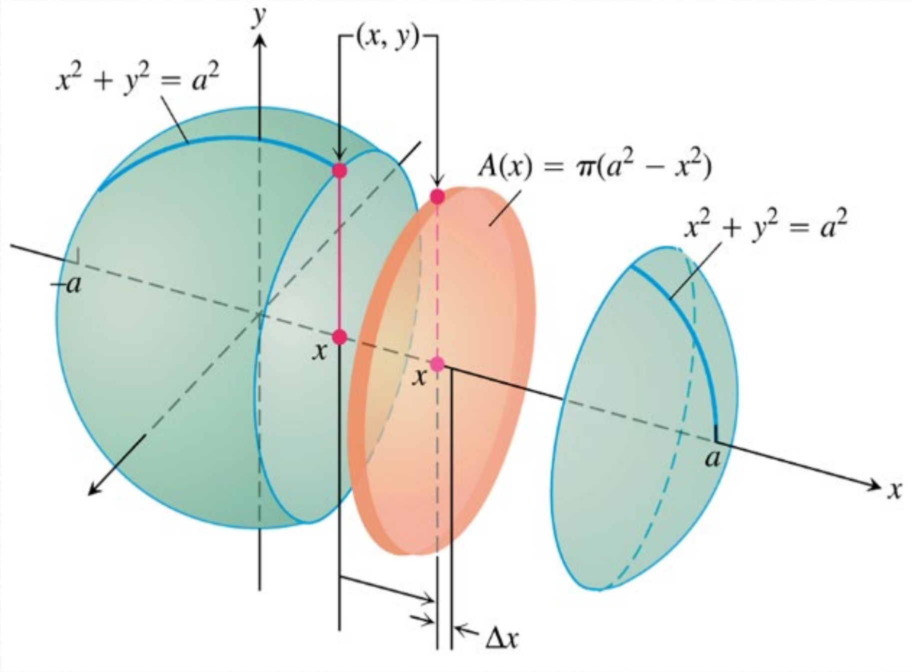
$$=\int_{-a}^{a}\pi(a^2-x^2)\,dx$$

$$=2\pi\int_0^a(a^2-x^2)\,dx$$

$$=2\pi\left(a^2x-\frac{1}{3}x^3\Big|_0^a\right)$$

$$=2\pi\left(a^3-\frac{1}{3}a^3\right)$$

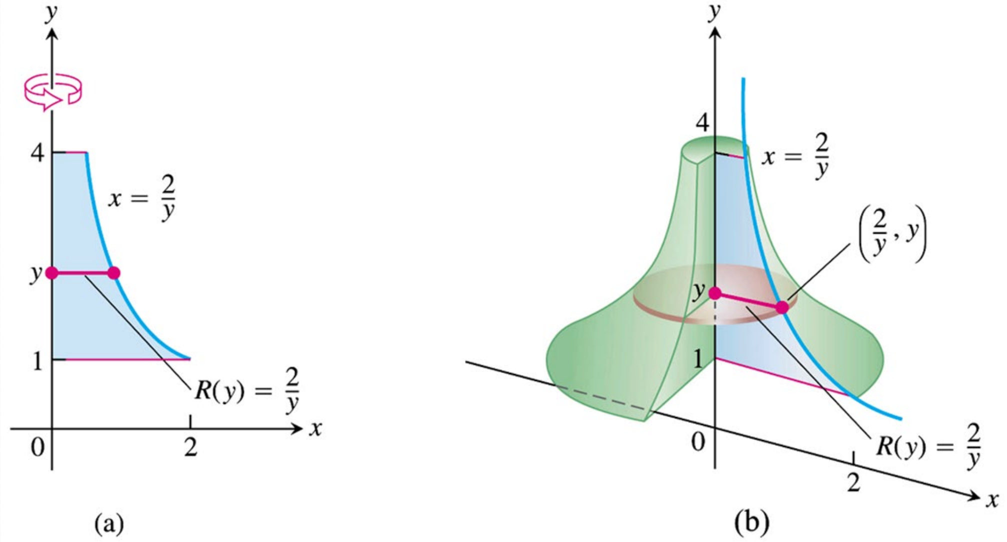
$$=\frac{4}{3}\pi a^3$$



## Volumes by Disks for Rotation About the $y-$Axis

$$V=\int_c^d A(y)\,dy=\int_a^b \pi\left[R(y)\right]^2 dy$$

Ex7(p379) Find the volume of the solid generated by revolving the region between the $y-$axis and the curve $x=\dfrac{2}{y}, 1\le y\le 4$, about the $y-$axis.




$$V=\int_1^4 \pi\left(\frac{2}{y}\right)^2 dy$$

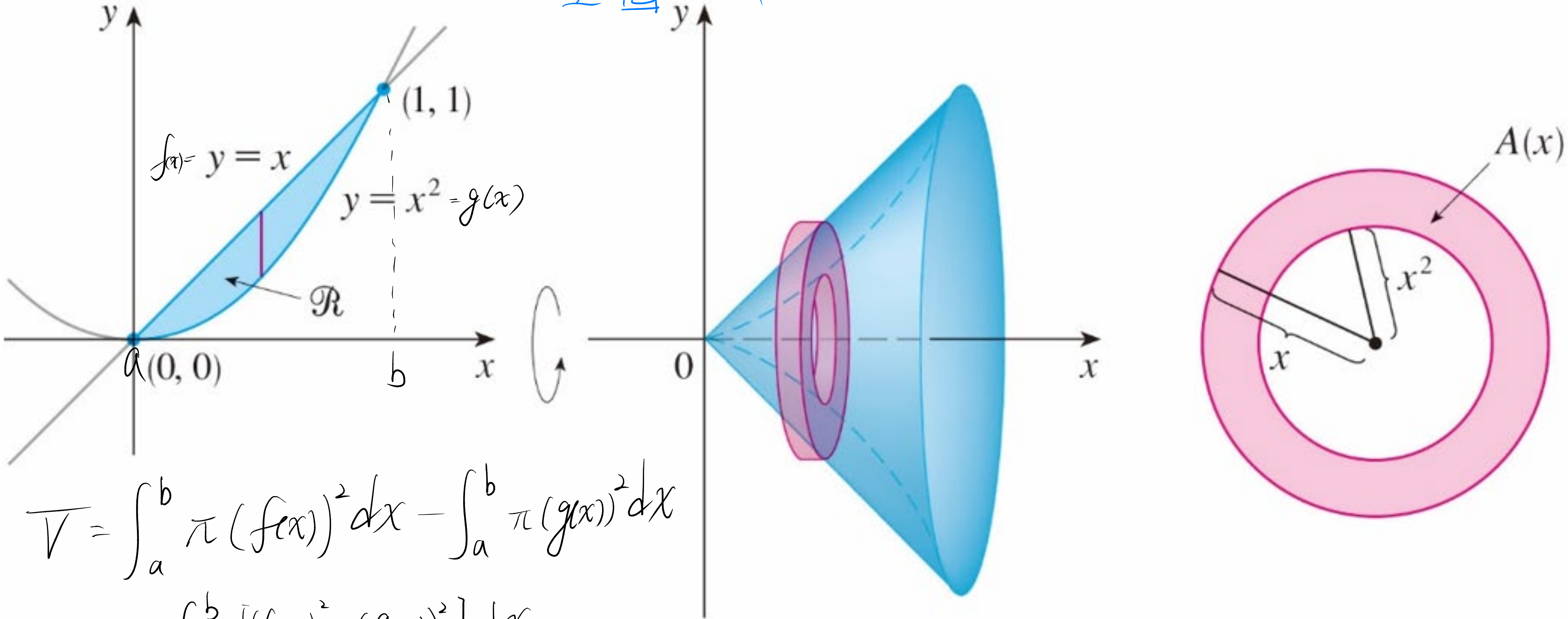
$$=\pi\int_1^4 \frac{4}{y^2}\,dy$$

$$=\pi\left(\frac{-4}{y}\Big|_1^4\right)$$

$$=\pi(-1+4)=3\pi$$

# Solid of Revolution: The Washer Method

垫圈法




$$V=\int_a^b \pi (f(x))^2\,dx-\int_a^b \pi (g(x))^2\,dx$$

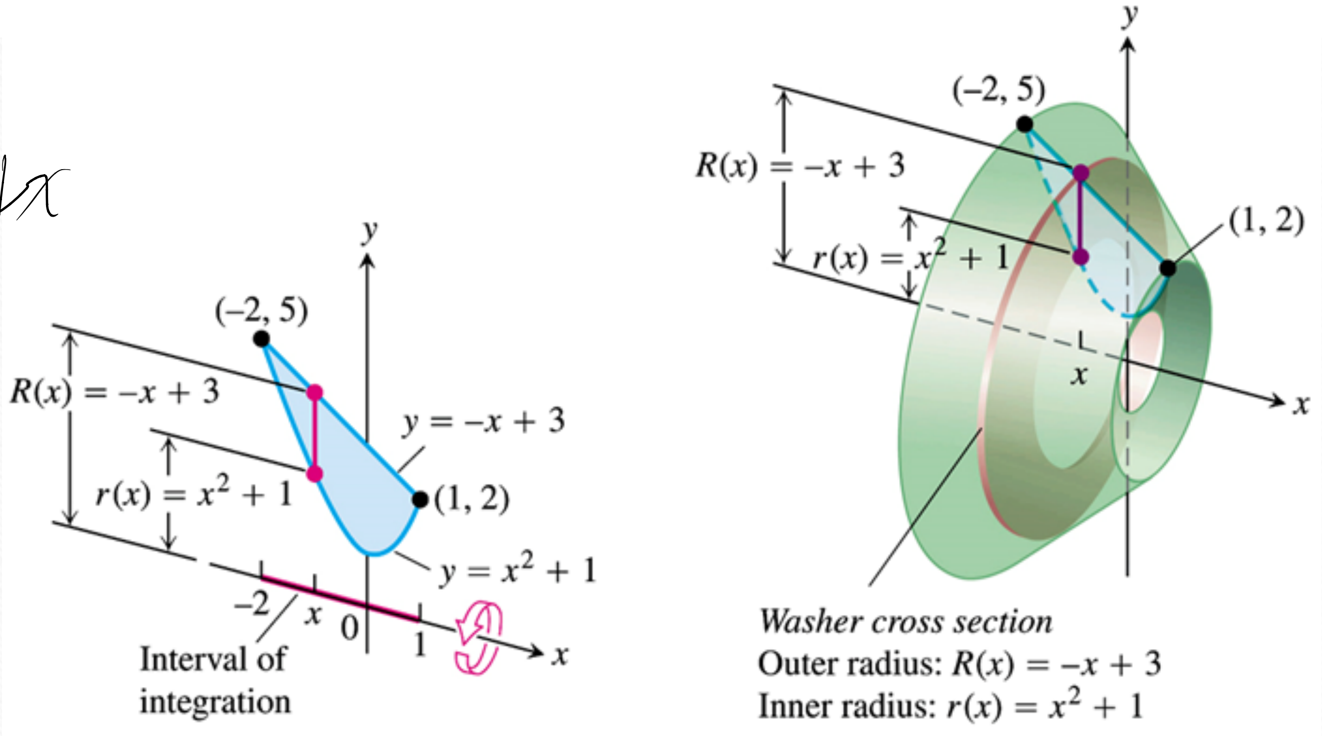
$$=\pi\int_a^b \left[(f(x))^2-(g(x))^2\right]dx$$

(a) (b) (c)

Ex9(p380) The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about the $x-$axis to generate a solid. Find the volume of the solid.

$$x^2+1=-x+3$$
$$x^2+x-2=0$$
$$x=-2,1$$

$$V=\int_{-2}^{1}\pi\left[(3-x)^2-(x^2+1)^2\right]dx$$
$$=\pi\int_{-2}^{1}\left[9-6x+x^2-(x^4+2x^2+1)\right]dx$$
$$=\pi\int_{-2}^{1}\left[9-6x+x^2-x^4-2x^2-1\right]dx$$
$$=\pi\int_{-2}^{1}(-x^4-x^2-6x+8)\,dx$$
$$=\pi\left(\frac{-x^5}{5}-\frac{x^3}{3}-3x^2+8x\Big|_{-2}^{1}\right)$$
$$=\pi\left[\frac{-1}{5}-\frac{1}{3}-3+8-\left(\frac{32}{5}+\frac{8}{3}-12-16\right)\right]$$
$$=\pi\left[-\frac{33}{5}-3+33\right]=\pi\left(\frac{117}{5}\right)$$




(a) (b)

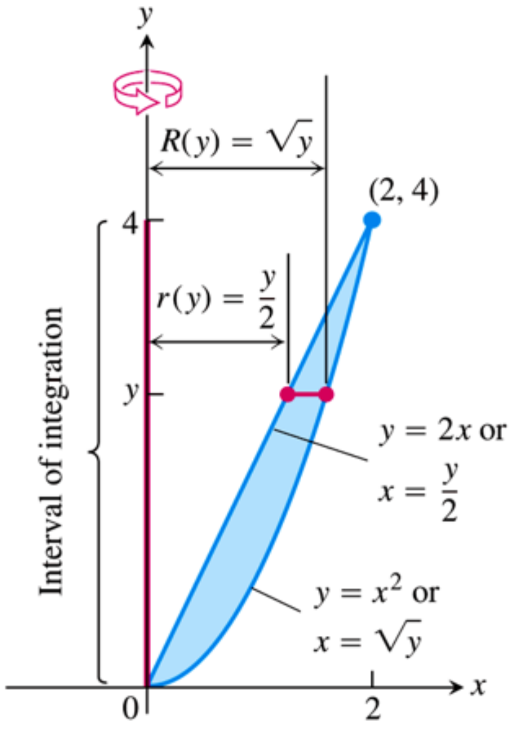
Ex10(p381) The region bounded by the parabola $y = x^2$ and the line $y = 2x$ in the first quadrant is revolved about the $y-$axis to generate a solid. Find the volume of the solid.

$$V = \pi \int_0^4 \left[ (\sqrt{y})^2 - \left(\frac{y}{2}\right)^2 \right] dy$$

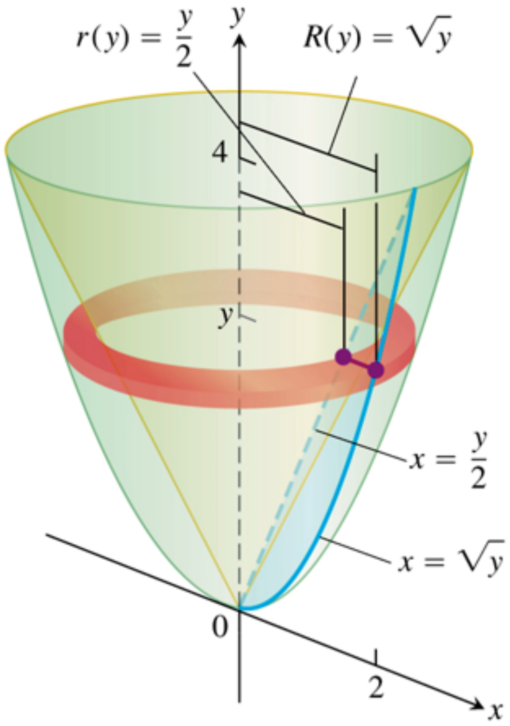
$$= \pi \int_0^4 \left( y - \frac{y^2}{4} \right) dy$$

$$= \pi \left( \frac{y^2}{2} - \frac{y^3}{12} \Big|_0^4 \right)$$

$$= \pi \left( \frac{4^2}{2} - \frac{4^3}{12} \right) = \frac{8}{3}\pi$$



(a)



(b)

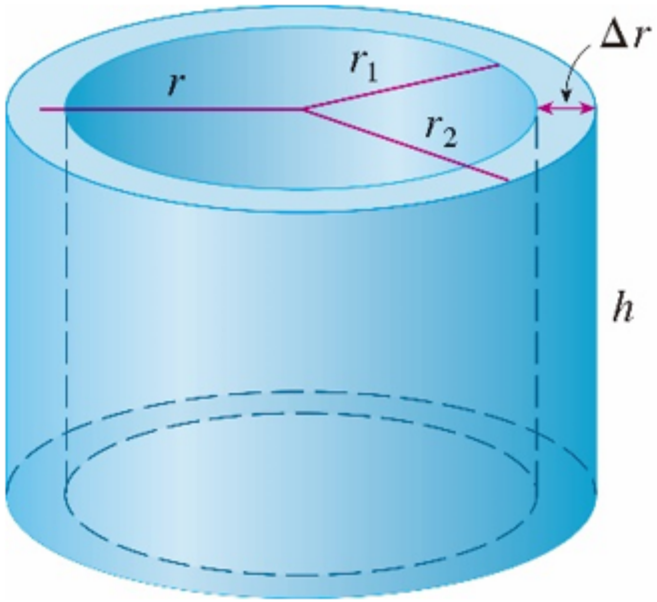
# HW6-1

- **HW: 2,5,10,11,18,26,28,29,37,38,50.**

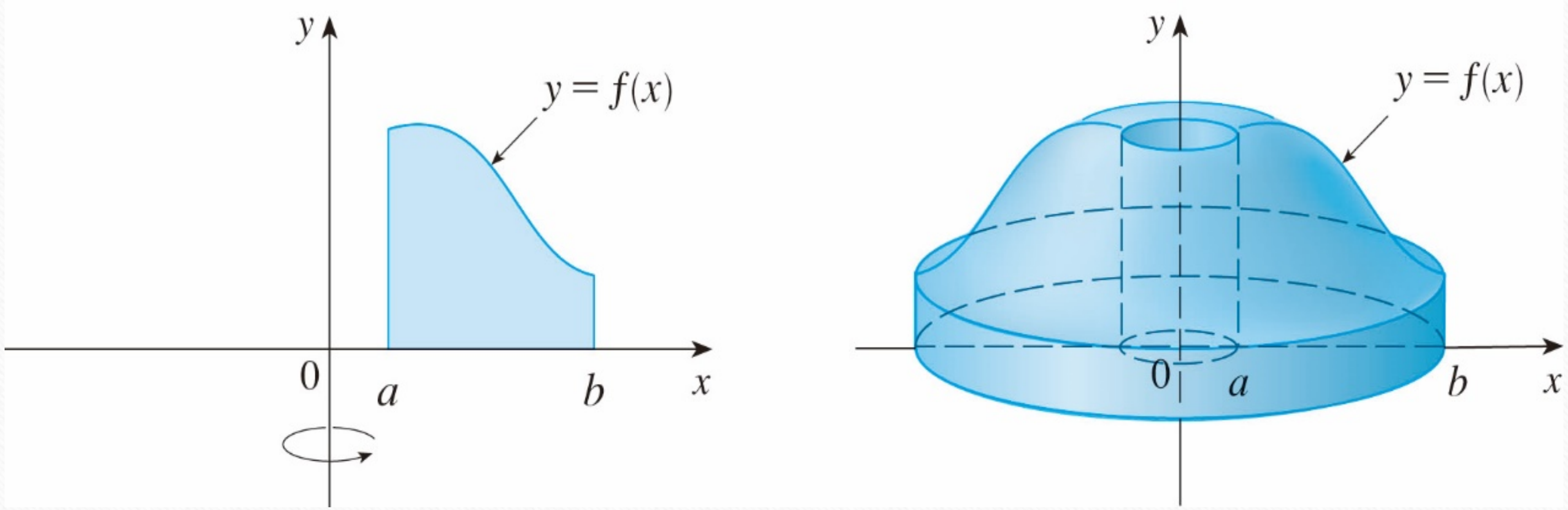
# 6-2 Volumes Using Cylindrical Shells

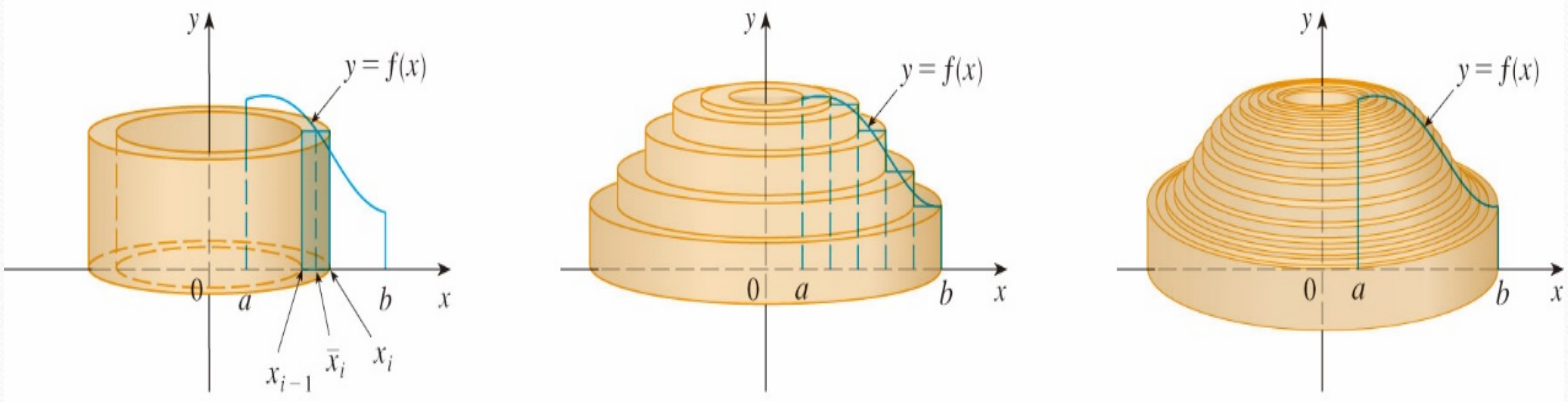
師大工教一

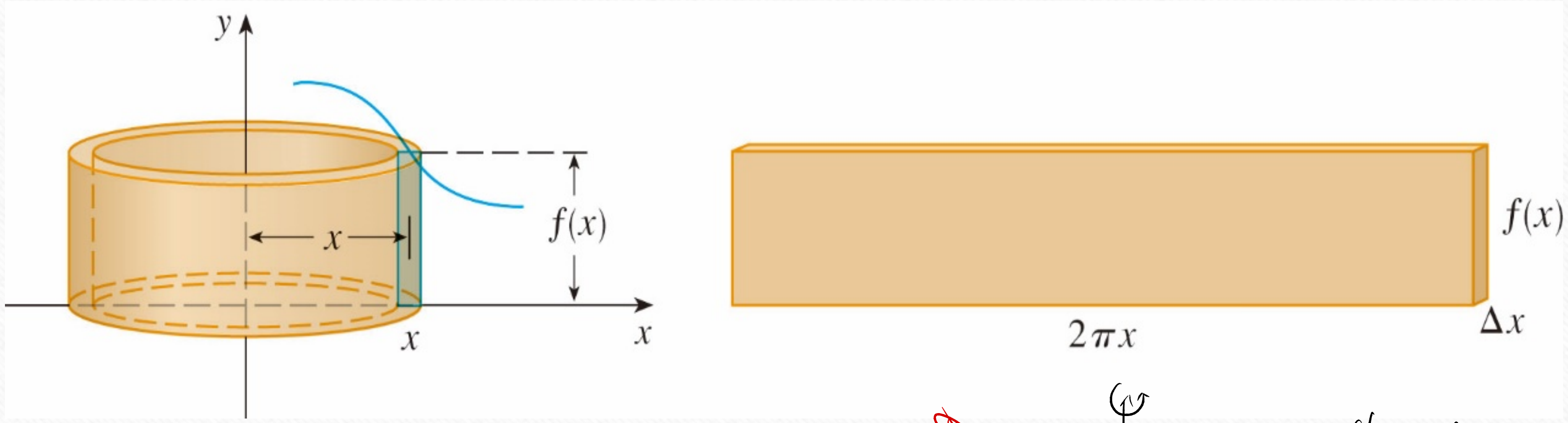
## Cylindrical Shell

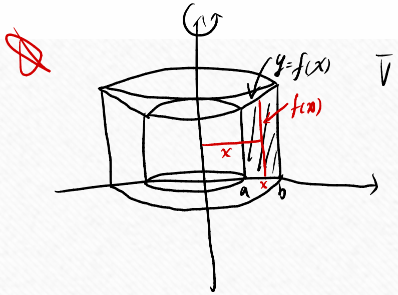



$$V = \pi r_2^{\,2} h - \pi r_1^{\,2} h = 2\pi\left(\frac{r_2 + r_1}{2}\right) h\left(r_2 - r_1\right) = 2\pi \cdot r \cdot h\Delta r$$


y
y = f(x)
0
a
b
x
y
y = f(x)
0
a
b
x


y
y = f(x)
0
a
b
x
$x_{i-1}$
$\bar{x}_i$
$x_i$
y
y = f(x)
0
a
b
x
y
y = f(x)
0
a
b
x



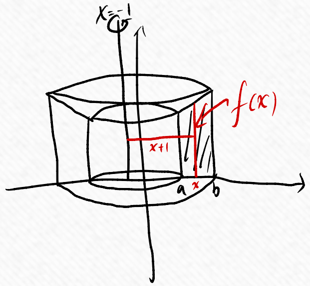




$$V = \int_a^b 2\pi x f(x)\,dx$$

Rotating about $y$ – axis

$$\Delta V_k = 2\pi x\, f(x)\,\Delta x_k$$

$$V = \lim_{n\to\infty}\sum_{k=1}^{n}\Delta V_k = \int_a^b 2\pi x\, f(x)\,dx$$



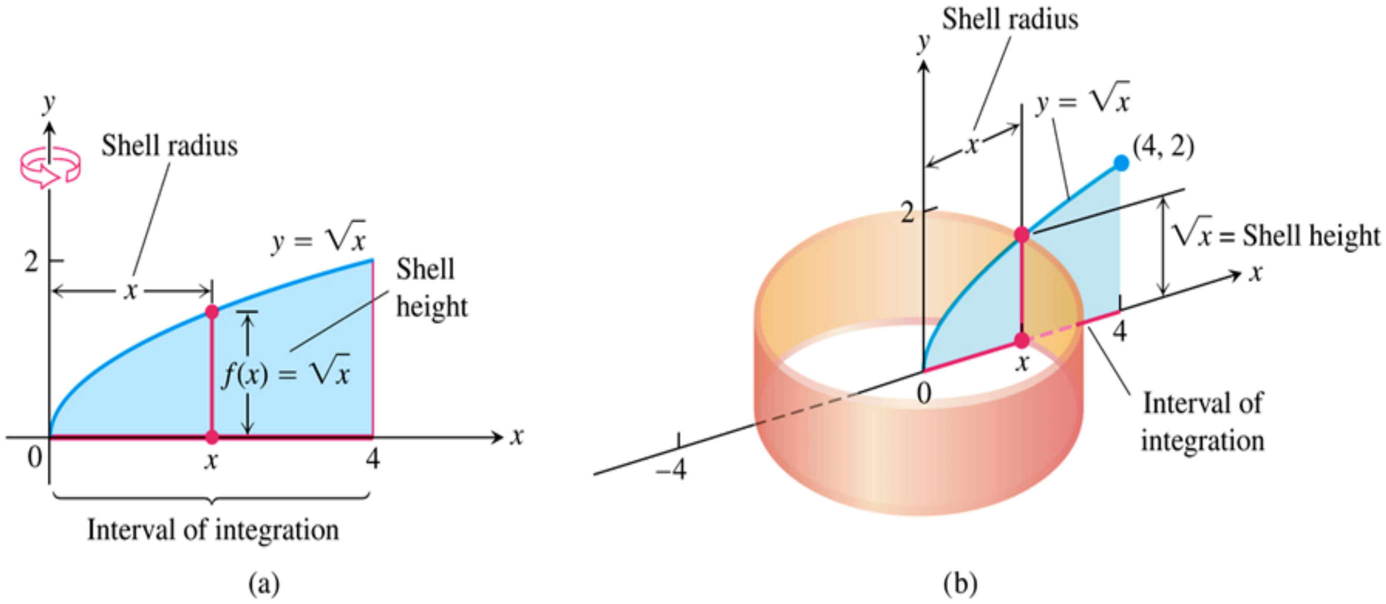

$$V = \int_a^b 2\pi(x+1) f(x)\,dx$$

Comparing the solid rotating about $y-$axis and $x=-1$

**Shell Formula for Revolution About a Vertical Line**

The volume of a solid generated by revolving the region between the $x-$axis and the graph of a continuous function $y=f(x)\geq 0, L\leq a\leq x\leq b$, about a vertical line $x=L$ is $V=\int_a^b 2\pi \begin{pmatrix}\text{shell}\\ \text{radius}\end{pmatrix}\begin{pmatrix}\text{shell}\\ \text{height}\end{pmatrix}dx$.

Ex2(p388) The region bounded by the curve $y=\sqrt{x}$, the $x-$axis, and the line $x=4$ is revolved about the $y-$axis to generate a solid. Find the volume of the solid.



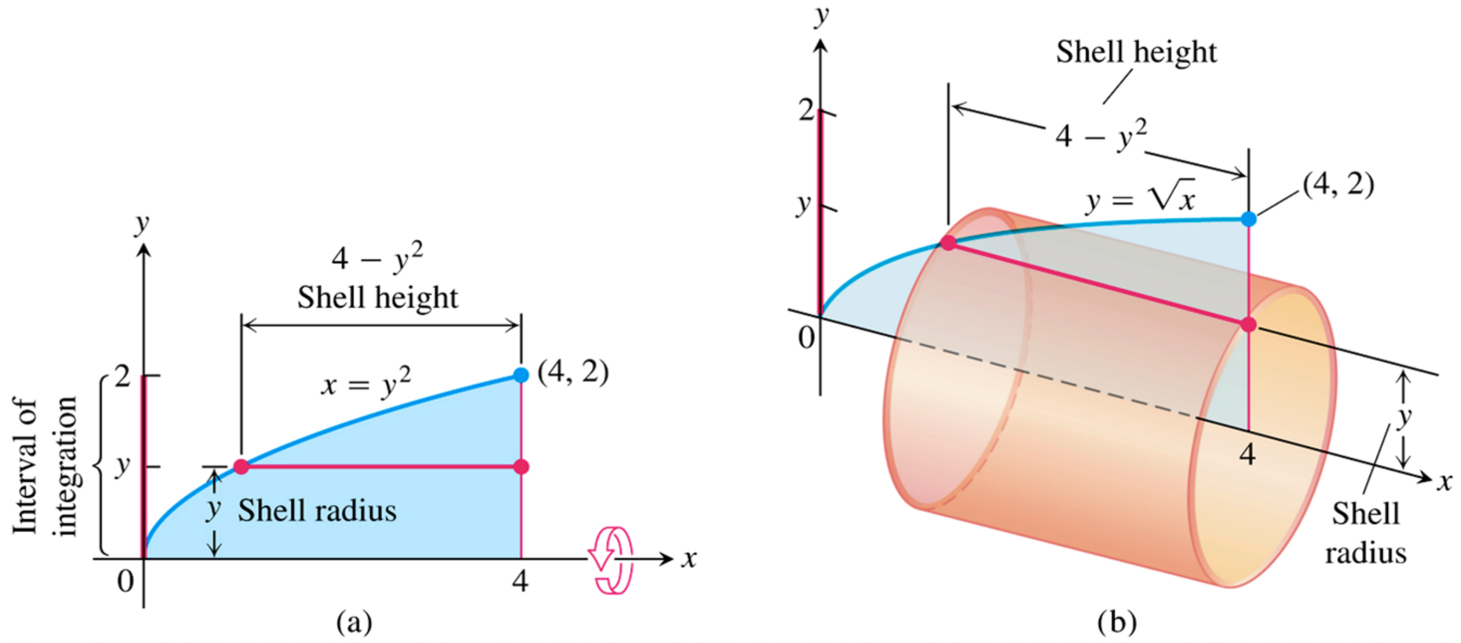

$$V=\int_0^4 2\pi(x)(\sqrt{x})\,dx$$

$$=\int_0^4 2\pi x^{\frac{3}{2}}\,dx$$

$$=2\pi\left(\frac{2}{5}x^{\frac{5}{2}}\Big|_0^4\right)$$

$$=\frac{128}{5}\pi$$

Ex3(p389) The region bounded by the curve $y=\sqrt{x}$, the $x-$axis, and the line $x=4$ is revolved about the $x-$axis to generate a solid. Find the volume of the solid.




$x = y^2$

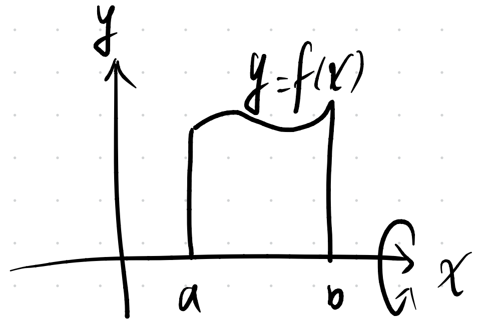
$V = \int_0^2 2\pi y(4-y^2)\,dy$

$= 2\pi \int_0^2 (4y - y^3)\,dy$

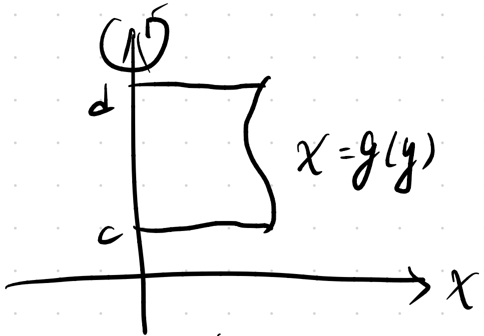
$= 2\pi \left(2y^2 - \frac{y^4}{4}\Big|_0^2\right)$

$= 2\pi(8-4) = 8\pi$

# Disk

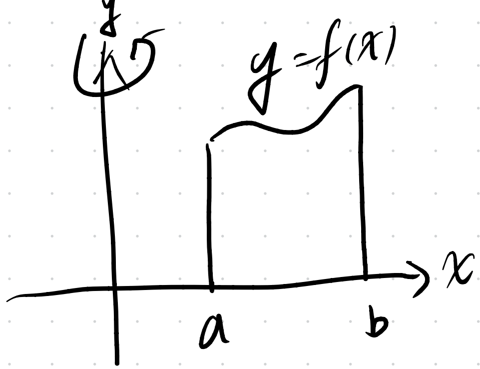



$$V=\int_a^b \pi [f(x)]^2 dx$$

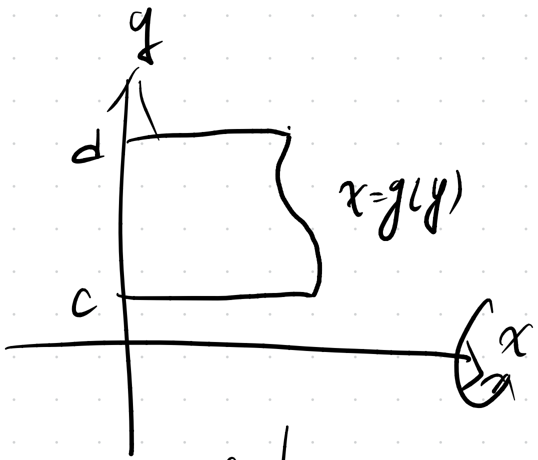



$$V=\int_c^d \pi (g(y))^2 dy$$

# Shell




$$V=\int_a^b 2\pi x f(x) dx$$




$$V=\int_c^d 2\pi y g(y) dy$$

**Summary of the Shell Method**

Regardless of the position of the axis of revolution (horizontal or vertical), the steps for implementing the shell method are these.

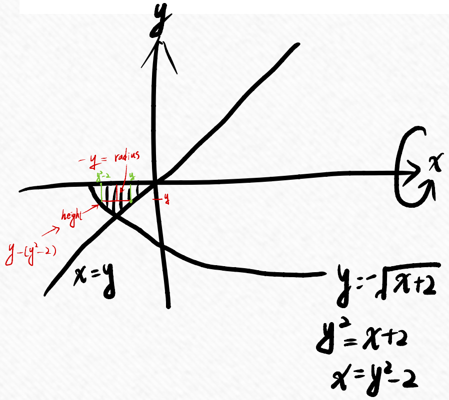
1. Draw the region and sketch a line segment across it parallel to the axis of revolution. Label the segment's height or length (shell height) and distance from the axis of revolution (shell radius).
2. Find the limit of integration for the thickness variable.
3. Integrate the product $2\pi$ (shell radius) (shell height) with respect to the thicness variable ($x$ or $y$) to find the volume.

Ex(107 年, #4) Let $D$ be the plane region bounded by

$$y=-\sqrt{x+2},\quad y=x,\quad y=0.$$

| $x$ | $-2$ | $-1$ | $2$ |
|---|---|---|---|
| $y$ | $0$ | $-1$ | $-2$ |

Plot the region $D$ and use **the shell method** to find the volume of the solid generated by revolving the region $D$ about the $x-$axis.




$$V=\int_{-1}^{0}2\pi(-y)(y-y^2+2)\,dy$$

$$=-2\pi\int_{-1}^{0}(y^2-y^3+2y)\,dy$$

$$=-2\pi\left(\frac{y^3}{3}-\frac{y^4}{4}+y^2\Big|_{-1}^{0}\right)$$

$$=+2\pi\left(\frac{-1}{3}-\frac{1}{4}+1\right)=2\pi\times\frac{5}{12}=\frac{5\pi}{6}$$ ☆

# HW6-2

- **HW: 4,5,9,13,16,25.**

105本 3. 給定由曲線 $y = x\sqrt{x+1}$ 與 $x$ 軸所圍成的區域，試做下列各題：

(1) 求該封閉區域的面積。(6 分)

(2) 求該區域對 $x$ 軸旋轉所得到的旋轉體體積。(6 分)

(3) 求該區域對 $y$ 軸旋轉所得到的旋轉體體積。(8 分)

(1) $A = -\int_{-1}^{0} \left(x\sqrt{x+1}\right)dx = -\int_{0}^{1} (u-1)\sqrt{u}\,du$

Let $u = x+1$

$du = dx$

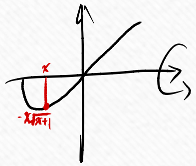
$x = u-1$

$= -\int_0^1 \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right)du$

$= -\left(\frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}}\Big|_0^1\right)$

$= -\left(\frac{2}{5} - \frac{2}{3}\right) = \frac{4}{15}$

(2)




$V = \int_{-1}^{0} \pi\left(-x\sqrt{x+1}\right)^2 dx$

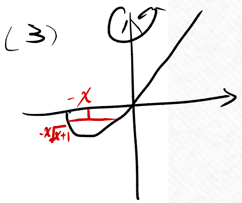
$= \pi\int_{-1}^{0} x^2(x+1)\,dx$

$= \pi\int_{-1}^{0} (x^3+x^2)\,dx$

$= \pi\left(\frac{1}{4}x^4 + \frac{1}{3}x^3\Big|_{-1}^{0}\right)$

$= \frac{\pi}{12}$

(3)




$V = \int_{-1}^{0} 2\pi(-x)\left(-x\sqrt{x+1}\right)dx$

$= 2\pi\int_{-1}^{0} x^2\sqrt{x+1}\,dx$

$= 2\pi\int_0^1 (u-1)^2\sqrt{u}\,du$

$= 2\pi\int_0^1 (u^2-2u+1)\sqrt{u}\,du$

$= 2\pi\int_0^1 \left(u^{\frac{5}{2}} - 2u^{\frac{3}{2}} + u^{\frac{1}{2}}\right)du$

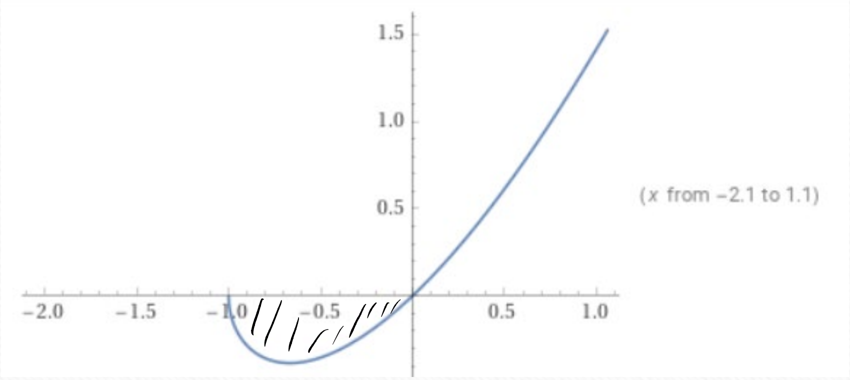
$= 2\pi\left(\frac{2}{7}u^{\frac{7}{2}} - \frac{4}{5}u^{\frac{5}{2}} + \frac{2}{3}u^{\frac{3}{2}}\Big|_0^1\right)$

$= 2\pi\left(\frac{2}{7} - \frac{4}{5} + \frac{2}{3}\right)$

$= \frac{32}{105}\pi$

plot | $y = x\sqrt{x+1}$

Plots



# 6-3 Arc Length

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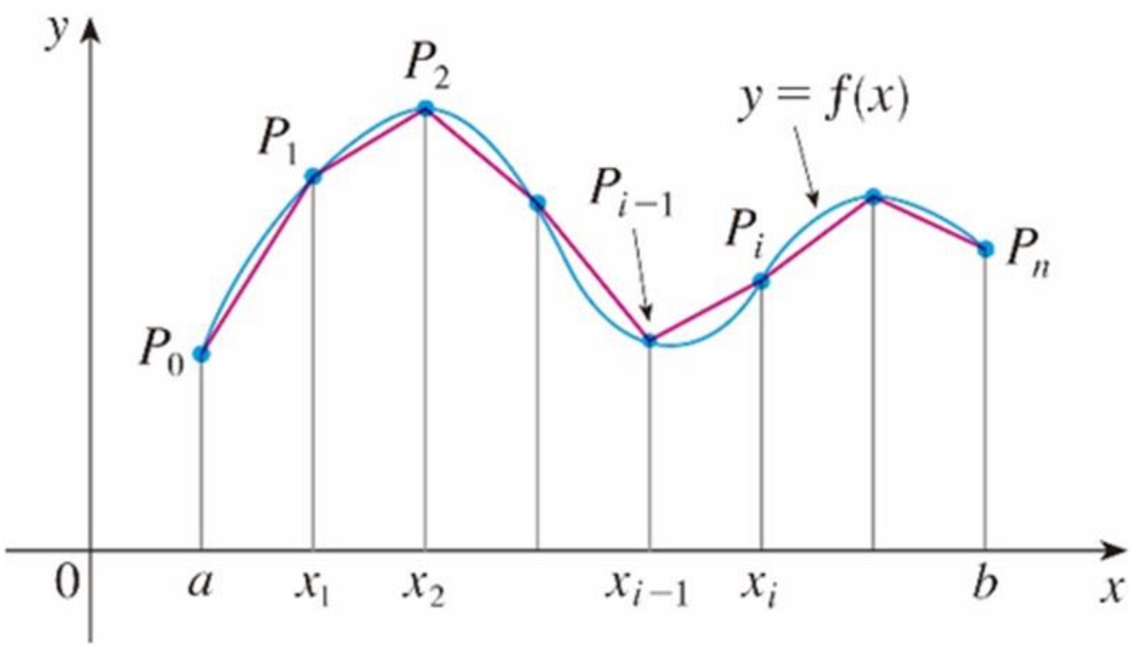
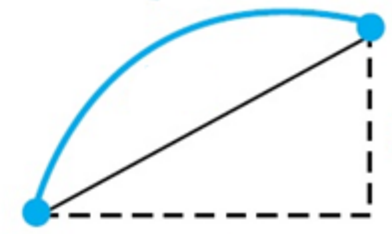
**Length of a curve** $y = f(x)$

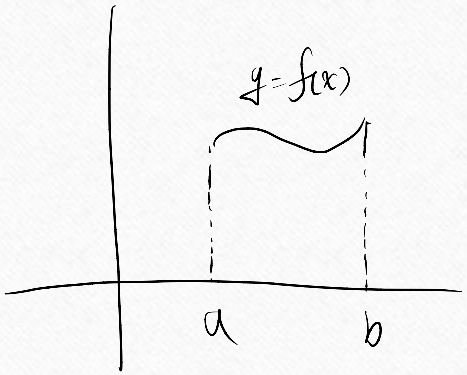


$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

$$\sum_{k=1}^{n} L_k = \sum_{k=1}^{n} \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = \sum_{k=1}^{n} \sqrt{(\Delta x_k)^2 + \left(f'(c_k)\Delta x_k\right)^2} = \sum_{k=1}^{n} \sqrt{1 + \left[f'(c_k)\right]^2}\,(\Delta x_k)$$

$$\lim_{n\to\infty} \sum_{k=1}^{n} L_k = \lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{1 + \left[f'(c_k)\right]^2}\,(\Delta x_k) = \int_a^b \sqrt{1 + \left[f'(x)\right]^2}\, dx$$

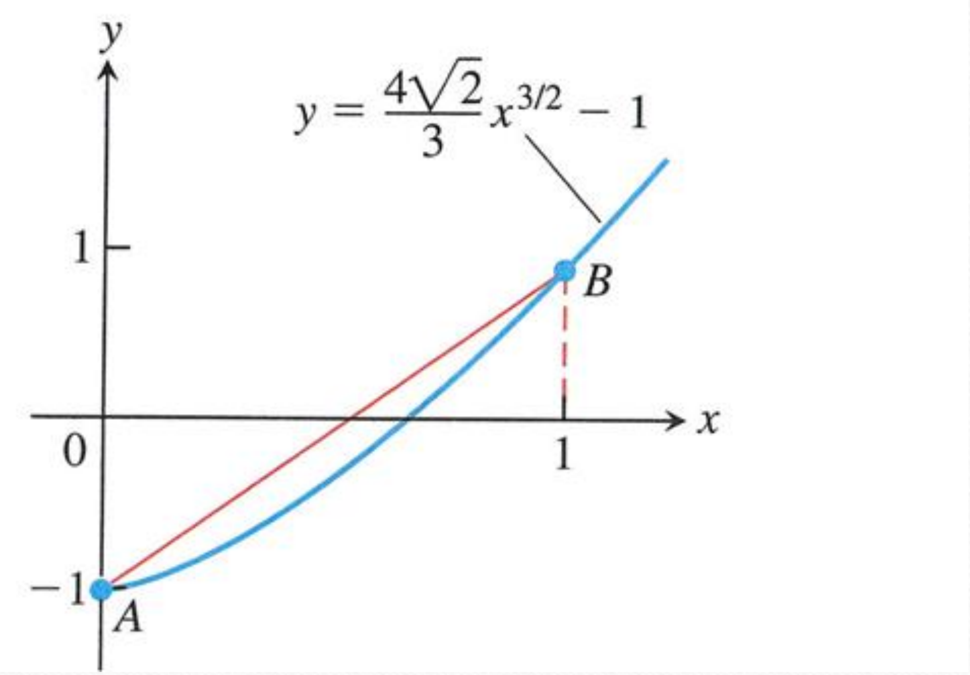



$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$$

Definition If $f'$ is continuous on $[a,b]$, then the **length** (**arc length**) of the curve $y = f(x)$ from the point $A = (a, f(a))$ to the point $B = (b, f(b))$ is the value of the integral $L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$.

Ex1(p394) Find the length of the curve shown in Figure 6.24, which is the graph of the function $y = \frac{4\sqrt{2}}{3} x^{\frac{3}{2}} - 1, \quad 0 \le x \le 1.$

$$\frac{dy}{dx} = 2\sqrt{2}\, x^{\frac{1}{2}} = 2\sqrt{2x}$$




$$L = \int_0^1 \sqrt{1+8x}\, dx$$

Let $u = 1+8x$
$du = 8dx$
$\frac{du}{8} = dx$

$$= \int_1^9 \sqrt{u}\, \frac{du}{8}$$

$$= \frac{1}{8}\left[\frac{2}{3} u^{\frac{3}{2}} \Big|_1^9\right] = \frac{1}{8}\left[\frac{2}{3}(1+8x)^{\frac{3}{2}} \Big|_0^1\right]$$

$$= \frac{1}{8} \cdot \left(\frac{2}{3} \times 27 - \frac{2}{3}\right) = \frac{1}{12} \times 26 = \frac{13}{6} \;\#$$

**Dealing with Discontinuities in $\dfrac{dy}{dx}$**

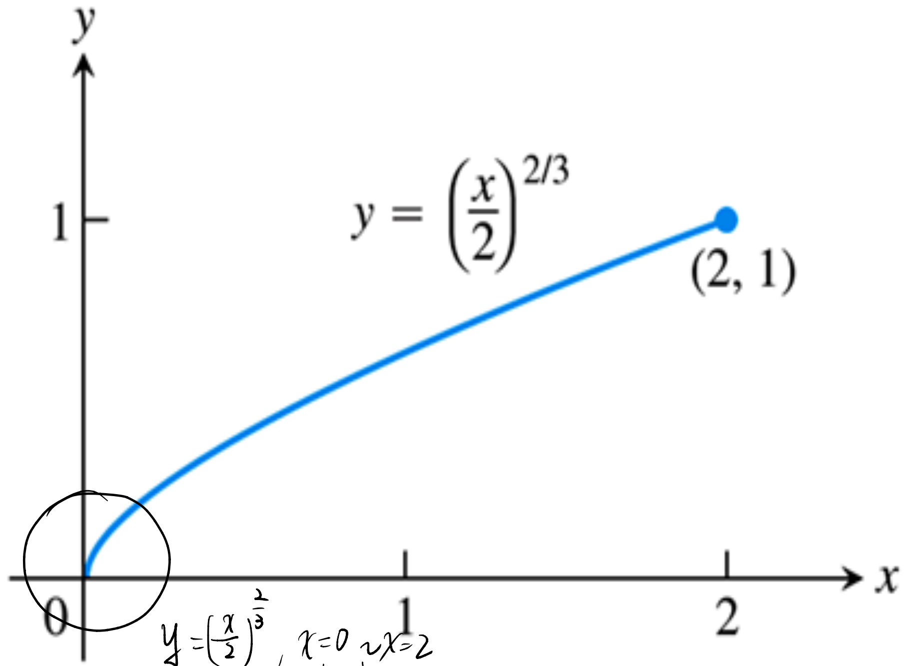
Formula for the length of $x = g(y), c \le y \le d$

If $g'$ is continuous on $[c, d]$, the length of the curve $x = g(y)$ from the point

$A = (g(c), c)$ to the point $B = (g(d), d)$ is

$$L = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy .$$

Ex4(p396) Find the length of the curve $y = \left(\frac{x}{2}\right)^{\frac{2}{3}}$ from $x = 0$ to $x = 2$.




$y = \left(\frac{x}{2}\right)^{\frac{2}{3}}$, $x=0 \sim x=2$

$y' = \frac{2}{3}\left(\frac{x}{2}\right)^{-\frac{1}{3}} \cdot \frac{1}{2}$

$y'\big|_{x=0}$ DNE.

$y^{\frac{3}{2}} = \frac{x}{2}$

$x = 2y^{\frac{3}{2}}$ $\quad x=0 \Rightarrow y=0$ $\quad \frac{dx}{dy} = 3y^{\frac{1}{2}}$

$x=2 \Rightarrow y=1$

Let $u = 1+9y$ $\quad du = 9dy$

$$L = \int_0^1 \sqrt{1+9y}\,dy = \int_1^{10} \sqrt{u}\,\frac{du}{9}$$

$$= \frac{2}{3} \times \frac{1}{9}\left[u^{\frac{3}{2}}\Big|_1^{10}\right]$$

$$= \frac{2}{27}\left(10^{\frac{3}{2}} - 1\right)$$

$\frac{dy}{dx} = \sqrt{\cos 4x}$

$\cos^2(2x) = \frac{1+\cos(4x)}{2}$

Ex(102 年, #3) 求曲線 $y = \int_0^x \sqrt{(\cos 4t)}\, dt, 0 \le x \le \frac{\pi}{4}$ 的長度。

$$L = \int_0^{\frac{\pi}{4}} \sqrt{1+\cos 4x}\, dx = \int_0^{\frac{\pi}{4}} \sqrt{2(\cos^2(2x))}\, dx = \sqrt{2}\int_0^{\frac{\pi}{4}} |\cos(2x)|\, dx = \sqrt{2}\cdot\frac{1}{2}\sin(2x)\Big|_0^{\frac{\pi}{4}} = \frac{1}{\sqrt{2}}(1-0) = \frac{1}{\sqrt{2}}$$

Ex(106 年, #4) Find the arc length of the graph of the function $y = \ln(\cos x)$ over $\left[0, \frac{\pi}{3}\right]$.

$$\frac{dy}{dx} = \frac{-\sin x}{\cos x} = -\tan x$$

Let $u = \sec x + \tan x$

$du = (\sec x \tan x + \sec^2 x)dx$

$= \sec x(\tan x + \sec x)dx$

$$L = \int_0^{\frac{\pi}{3}} \sqrt{1+\tan^2 x}\, dx$$

$$= \int_0^{\frac{\pi}{3}} \sec x\, dx$$

$$= \int_0^{\frac{\pi}{3}} \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x}\, dx$$

$$= \int_1^{2+\sqrt{3}} \frac{1}{u}\, du = \ln|u|\Big|_1^{2+\sqrt{3}} = \ln(2+\sqrt{3})$$

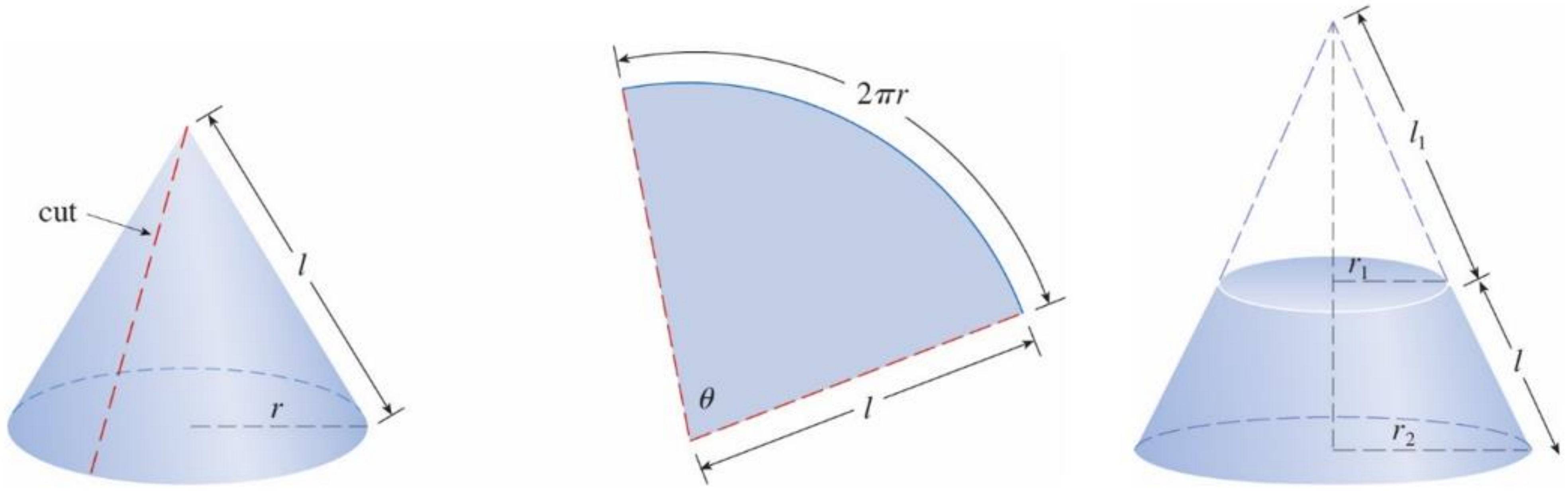
# HW6-3

- **HW: 2,4,15,16**

旋転表面積

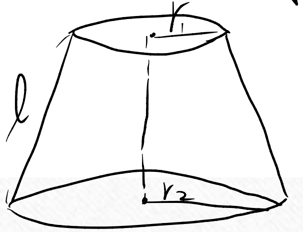
# 6-4 Areas of Surfaces of Revolution

師大工教一

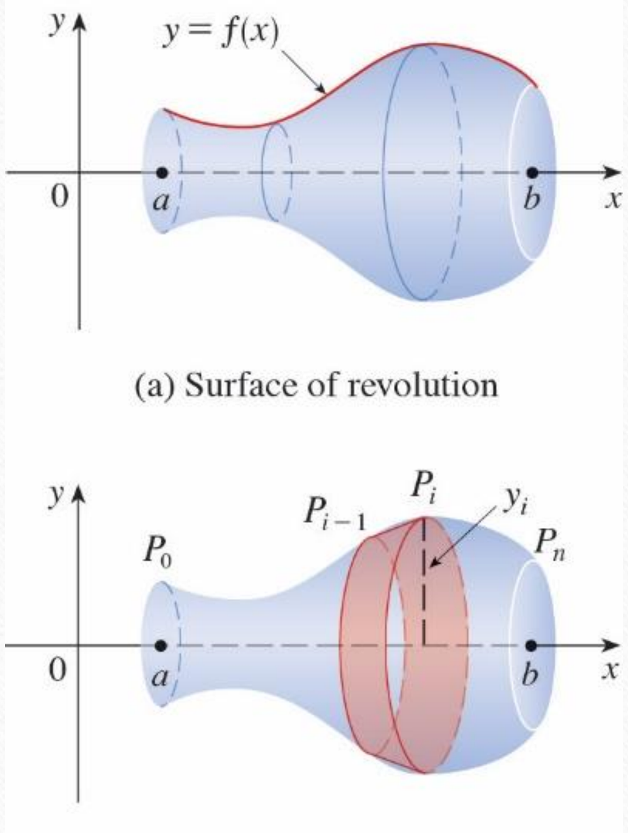



The area of surface of the frustum of a cone is $2\pi\frac{r_1+r_2}{2}l$.

圆锥台




$2\pi\left(\frac{r_1+r_2}{2}\right)l$



(a) Surface of revolution

(b) Approximating band

(1 piece)Frustum surface area

$$= 2\pi \cdot \frac{f\left(x_{k-1}\right) + f\left(x_k\right)}{2} \sqrt{\left(\Delta x_k\right)^2 + \left(\Delta y_k\right)^2}$$

$$\text{(n pieces) surface area}\begin{cases} =\sum_{k=1}^{n} 2\pi \cdot \dfrac{f\left(x_{k-1}\right)+f\left(x_k\right)}{2}\sqrt{\left(\Delta x_k\right)^2+\left(\Delta y_k\right)^2} \\ =\sum_{k=1}^{n} 2\pi \cdot \dfrac{f\left(x_{k-1}\right)+f\left(x_k\right)}{2}\sqrt{\left(\Delta x_k\right)^2+\left(f'\left(c_k\right)\Delta x_k\right)^2} \\ =\sum_{k=1}^{n} 2\pi \cdot \dfrac{f\left(x_{k-1}\right)+f\left(x_k\right)}{2}\sqrt{1+\left(f'\left(c_k\right)\right)^2}\,\Delta x_k \end{cases}$$

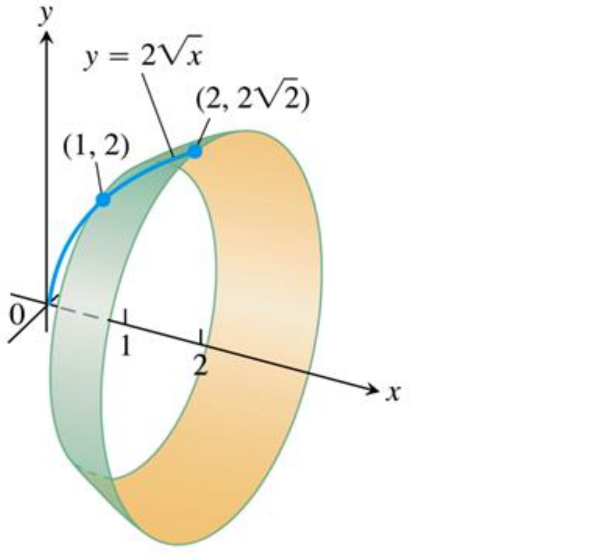
$$\text{Surface area}\begin{cases} =\lim\limits_{n\to\infty}\sum_{k=1}^{n} 2\pi \cdot \dfrac{f\left(x_{k-1}\right)+f\left(x_k\right)}{2}\sqrt{1+\left(f'\left(c_k\right)\right)^2}\,\Delta x_k \\ =\int_a^b 2\pi f(x)\sqrt{1+\left(f'(x)\right)^2}\,dx \end{cases}$$

Definition If the function $f(x) \geq 0$ is continuously differentiable on $[a,b]$, the area of surface generated by revolving the graph of $y = f(x)$ about the $x-$ axis is $S = \int_a^b 2\pi y \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx = \boxed{\int_a^b 2\pi f(x)\sqrt{1+\left(f'(x)\right)^2}\, dx}$

Ex1(p401) Find the area of surface generated by revolving the curve

$y = 2\sqrt{x}$ , $1 \le x \le 2$, about the $x$ – axis.

$$\frac{dy}{dx} = \frac{1}{\sqrt{x}}$$




$$S = \int_1^2 2\pi (2\sqrt{x}) \sqrt{1 + \frac{1}{x}}\, dx$$

$$= 4\pi \int_1^2 \sqrt{x+1}\, dx$$

Let $u = x+1$

$du = dx$

$$= 4\pi \int_2^3 \sqrt{u}\, du$$

$$= 4\pi \left[ \frac{2}{3} u^{\frac{3}{2}} \Big|_2^3 \right]$$

$$= \frac{8}{3}\pi \left( 3^{\frac{3}{2}} - 2^{\frac{3}{2}} \right)$$

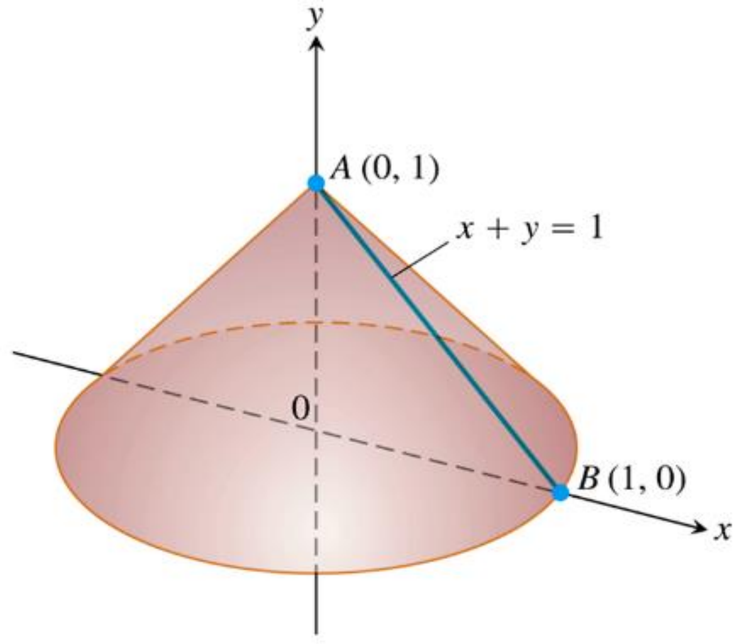
**Surface Area for Revolution About the $y-$Axis**

If the function $x=g(y)\geq 0$ is continuously differentiable on $[c,d]$, the area of surface generated by revolving the graph of $x=g(y)$ about the $y-$axis is

$$S=\int_c^d 2\pi x\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy=\int_a^b 2\pi g(y)\sqrt{1+\left(g'(y)\right)^2}\,dy$$

Ex2(p402) The line segment $x=1-y$, $0\le y\le 1$, is revolved about the $y-$axis to generate a cone. Find its lateral surface area(which excludes the base area).

$$\frac{dx}{dy}=-1$$




$$S=\int_0^1 2\pi(1-y)\sqrt{1+1}\,dy$$

$$=2\sqrt{2}\pi\int_0^1 (1-y)\,dy$$

$$=2\sqrt{2}\pi\left[y-\frac{y^2}{2}\Big|_0^1\right]$$

$$=2\sqrt{2}\pi\left(1-\frac{1}{2}\right)=\pi\sqrt{2}\ \#$$

# HW6-4

- **HW: 4,7,9,11,19,29**

(105分部) 4 (共 14 分) Let A be the curve $y = \sqrt{9 - x^2}$, $-1 \le x \le 1$.

(1). (8 分) Find the arc length of the curve A.

(2). (6 分) Find the area of the surface generated by revolving the curve A about the x-axis.

$$\frac{dy}{dx} = \frac{1}{\cancel{2}}(9-x^2)^{-\frac{1}{2}} \cdot (-\cancel{2}x) = \frac{-x}{\sqrt{9-x^2}}$$

(1)

$$L = \int_{-1}^{1} \sqrt{1 + \frac{x^2}{9-x^2}}\, dx$$

(2)

$$S = \int_{-1}^{1} 2\pi \sqrt{9-x^2} \sqrt{1+\frac{x^2}{9-x^2}}\, dx$$

$$= 2\pi \int_{-1}^{1} \sqrt{9 \cancel{-x^2+x^2}}\, dx = 2\pi \int_{-1}^{1} 3\, dx$$

$$= 6\pi \int_{-1}^{1} dx = 6\pi \left(x \Big|_{-1}^{1}\right) = 12\pi$$